

CS-4053 Recommender System

Lecture 7: Matrix Factorization

Flow of this lecture

- ☐ Features
- ☐ Factorizing interaction matrix
- ☐ Storage
- ☐ Optimization
- ☐ Predictions

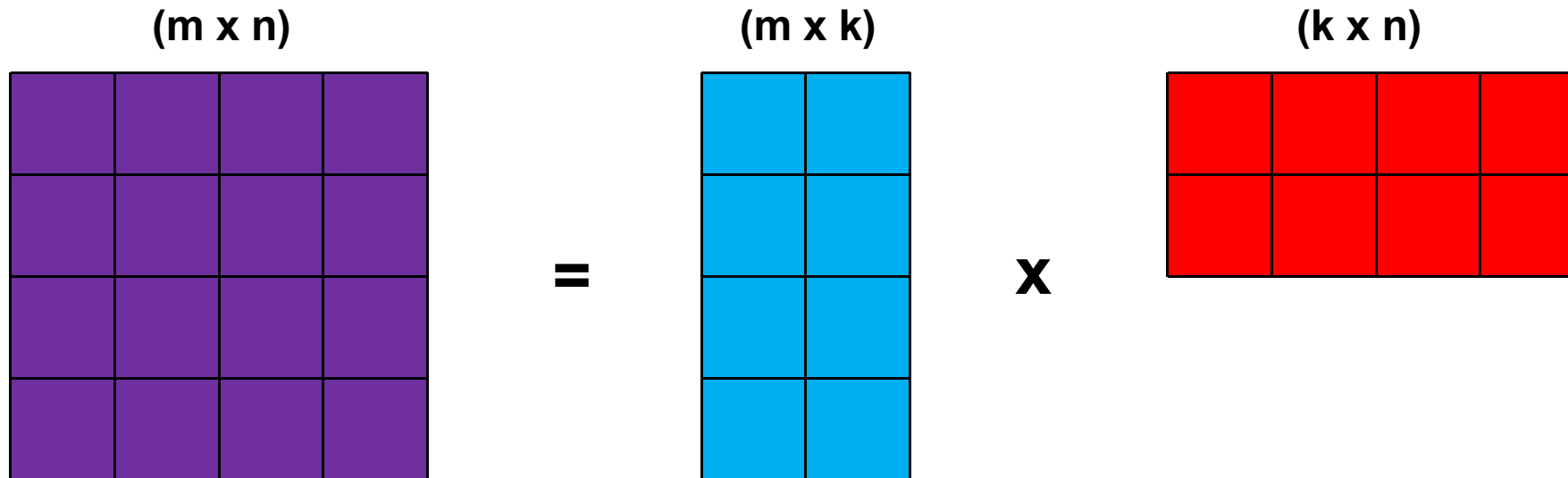
Factorization

- ❑ **Factorization** is a mathematical technique that allows a term to be decomposed into a product of two or more smaller terms

$$16 = 8 \times 2$$

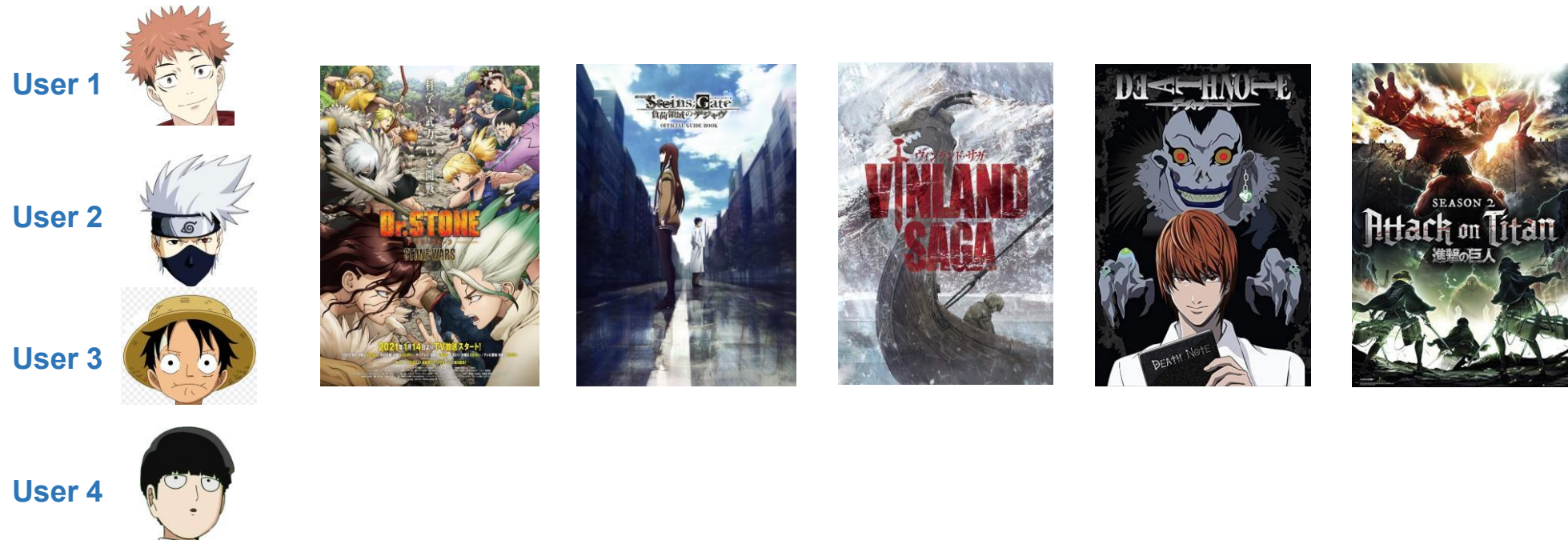
Matrix Factorization

- ❑ **Matrix Factorization** is a technique in which user-item interaction matrix is decomposed into a product of two or three matrices



Matrix Factorization

- We have a set of users and their interaction matrix for anime



Matrix Factorization

						
		I1	I2	I3	I4	I5
User 1		2	1	4	1	2
User 2		1	4	1	2	2
User 3		3	5	5	3	4
User 4		2	3	?	2	3

Matrix Factorization

- Let us consider some anime, their features, and **User 1**

User 1 

Action	Sci-fi
1	0



Action	Sci-fi
2	1



Action	Sci-fi
1	4



Action	Sci-fi
4	1

Matrix Factorization

- Let us consider some anime, their features, and **User 1**

User 1



Action	Sci-fi
1	0

$$1 \times 2 + 0 \times 1 = 2$$

(User 1's rating for I1)



Action	Sci-fi
2	1



Action	Sci-fi
1	4



Action	Sci-fi
4	1

Matrix Factorization

□ Let us try this with **User 2**

User 2



Action	Sci-fi
0	1

$$0 \times 2 + 1 \times 1 = 1$$

(User 2's rating for I1)



Action	Sci-fi
2	1



Action	Sci-fi
1	4



Action	Sci-fi
4	1

Matrix Factorization



	F1	F2
User 1 	1	0
User 2 	0	1
User 3 	1	1
User 4 	1	0.5

(m x k)

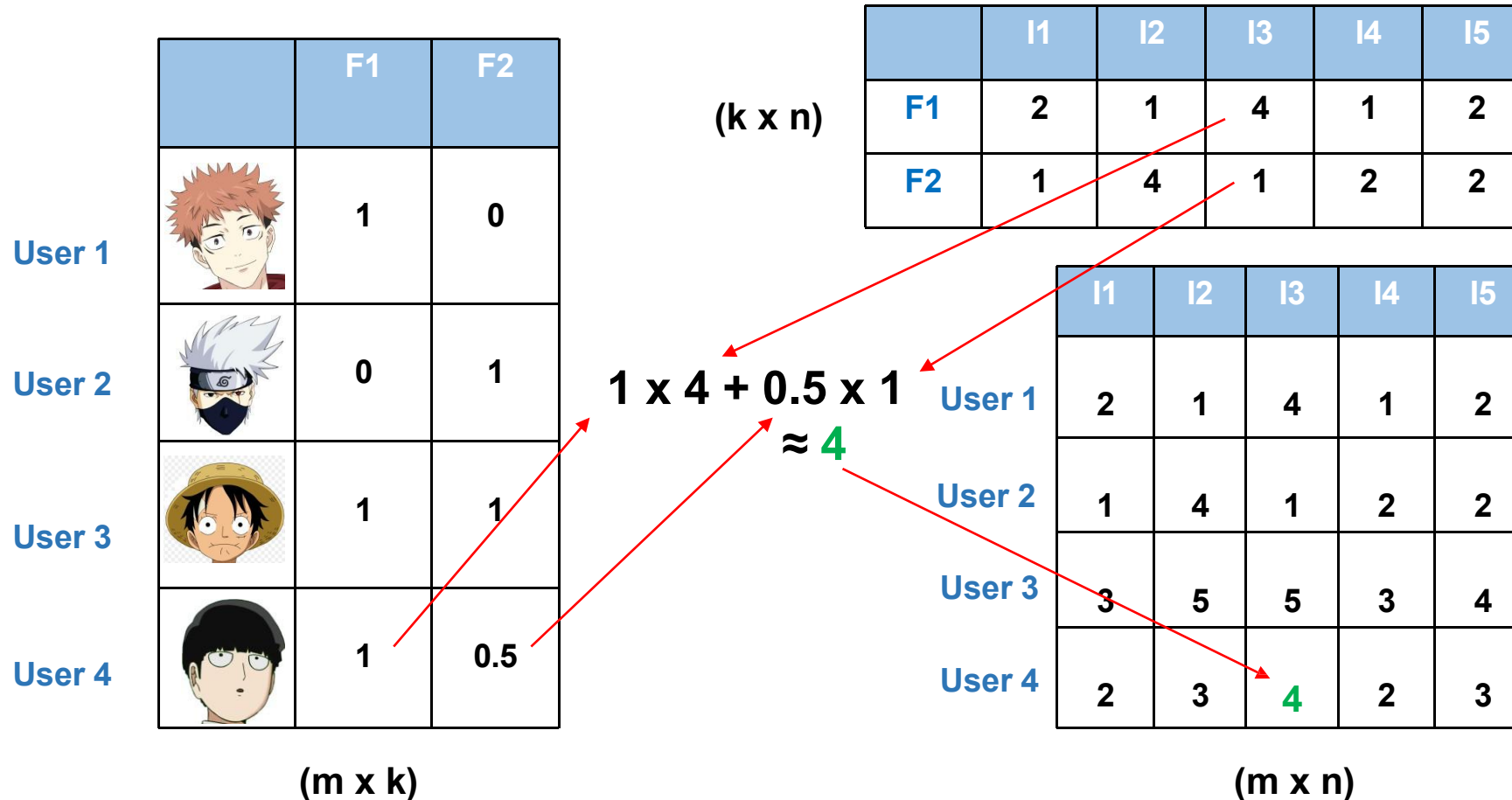
(k x n)

	I1	I2	I3	I4	I5
F1	2	1	4	1	2
F2	1	4	1	2	2

	I1	I2	I3	I4	I5
User 1	2	1	4	1	2
User 2	1	4	1	2	2
User 3	3	5	5	3	4
User 4	2	3	?	2	3

(m x n)

Matrix Factorization: Prediction



Matrix Factorization: Finding Factors

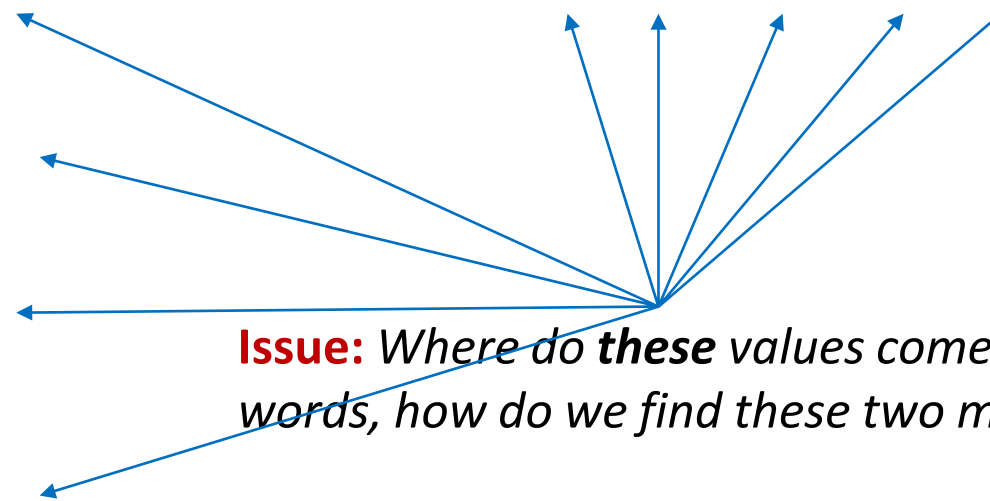


	F1	F2
User 1 	1	0
User 2 	0	1
User 3 	1	1
User 4 	1	0.5

(m x k)

(k x n)

	I1	I2	I3	I4	I5
F1	2	1	4	1	2
F2	1	4	1	2	2



Issue: Where do *these* values come from? In other words, how do we find these two matrices (factors)?

Matrix Factorization: Finding Factors

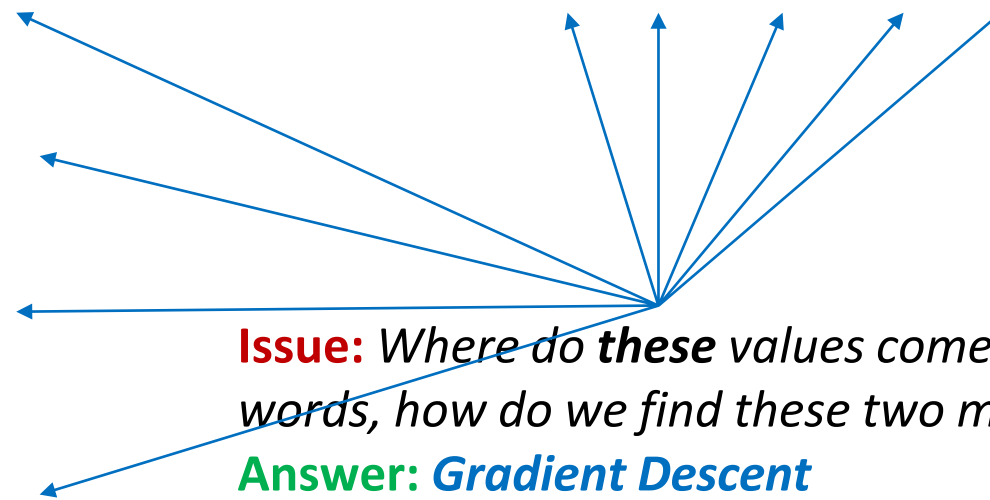


	F1	F2
User 1 	1	0
User 2 	0	1
User 3 	1	1
User 4 	1	0.5

(m x k)

(k x n)

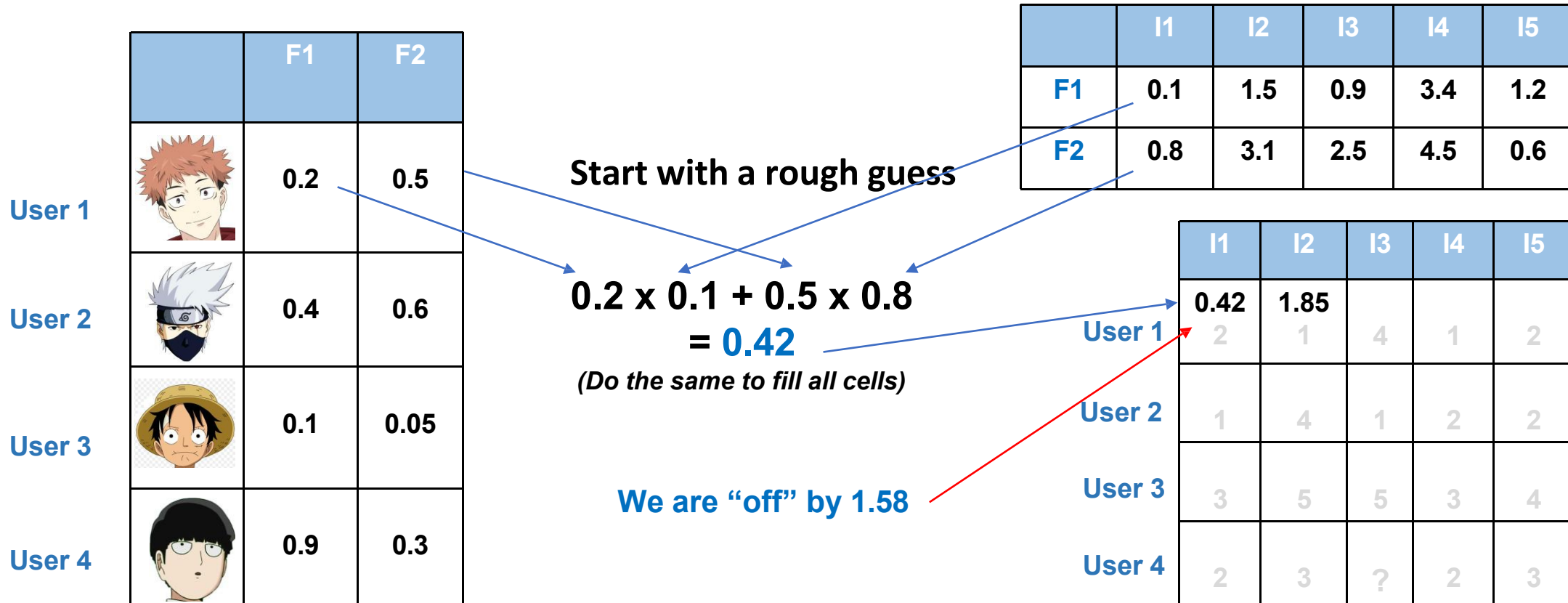
	I1	I2	I3	I4	I5
F1	2	1	4	1	2
F2	1	4	1	2	2



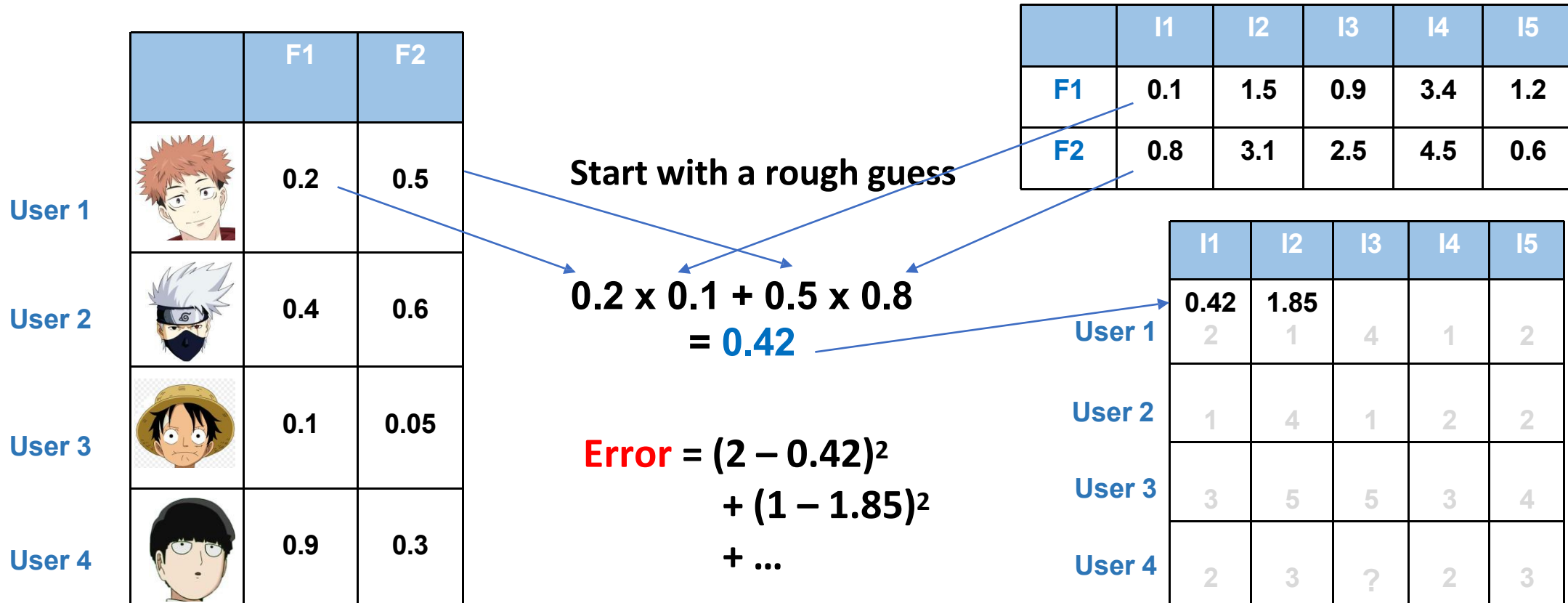
Issue: Where do *these* values come from? In other words, how do we find these two matrices (factors)?

Answer: Gradient Descent

Matrix Factorization: Gradient Descent



Matrix Factorization: Gradient Descent



Matrix Factorization: Gradient Descent



		F1	F2
User 1		0.14	0.45
User 2		0.32	0.5
User 3		0.43	0.07
User 4		0.8	0.3

Let's try this again with different values

$$0.14 \times 0.2 + 0.45 \times 1.1 = 0.52$$

Keep repeating until error can be minimized

	I1	I2	I3	I4	I5
F1	0.2	1.7	0.8	3.7	1.9
F2	1.1	3.3	3.2	4.0	0.5

	I1	I2	I3	I4	I5
User 1	0.52 2	1.72 1	4	1	2
User 2	1	4	1	2	2
User 3	3	5	5	3	4
User 4	2	3	?	2	3

Matrix Factorization: Finding Factors



	F1	F2
User 1 	1	0
User 2 	0	1
User 3 	1	1
User 4 	1	0.5

(m x k)

(k x n)

	I1	I2	I3	I4	I5
F1	2	1	4	1	2
F2	1	4	1	2	2

To find these values we can also use:

- **Genetic Algorithm**
- **Linear Programming**
- **PSO**
any other optimization technique

Matrix Factorization: Pros and Cons

Pros

- It can take much less storage e.g. a **2000x1000** interaction matrix can be stored as two matrices of **2000x100** and **100x1000** dimensions
Storage taken by 2000x1000 matrix = **2M**
Storage taken by separate matrices = **200k + 100k i.e., 300k**
- Predictions can be calculated quickly and easily

Cons

- The cost of optimization during training is non-deterministic